MonoArt: Progressive Structural Reasoning for Monocular Articulated 3D Reconstruction

Haitian Li, Haozhe Xie, Junxiang Xu, Beichen Wen, Fangzhou Hong, Ziwei Liu | MMLab@NTU

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Research Questions

- How can we reconstruct articulated 3D objects from a single image without relying on multi-view supervision, retrieval databases, or auxiliary video generation?
- How can progressive structural reasoning enable stable and interpretable articulation inference by disentangling geometry, part structure, and motion?
- Can a unified end-to-end framework achieve both high reconstruction accuracy and fast inference for monocular articulated 3D reconstruction?

Core Idea

The core problem is that directly regressing articulation parameters from image features is unstable due to entanglement between motion cues and object structure. MonoArt's key insight is to use progressive structural reasoning - first reconstruct canonical geometry using TRELLIS, then build part-aware representations through triplane projection and transformers, then use dual queries to decouple spatial motion anchors from semantic part representations, and finally predict kinematic parameters. This staged approach with explicit 3D structural priors enables stable articulation inference without external motion templates.

Key Formula

Pairwise Semantic Attachment Score

$$\mathbf{S}_{i,j} = \mathbf{s}_i^\top \mathbf{C} \mathbf{s}_j$$

Computes semantic attachment compatibility between parts i and j using learnable matrix C ∈ R^{Nc×Nc}, where si and sj are predicted category distributions. This bilinear formulation captures category-level attachment priors for kinematic tree construction.

Part-Aware Triplet Loss

$$\ell_{\text{triplet}} = -\frac{1}{2}\left[\log \frac{s_{ab}}{s_{ab}+s_{ac}} + \log \frac{s_{ab}}{s_{ab}+s_{bc}}\right]$$

Supervises part-aware semantic reasoner with triplet (ha,hb,hc) where (ha,hb) belong to same articulated part and hc to different part. The similarity scores s_ij use cosine similarity with learnable temperature. This enforces discriminative motion-aware part embeddings.

Key Figure

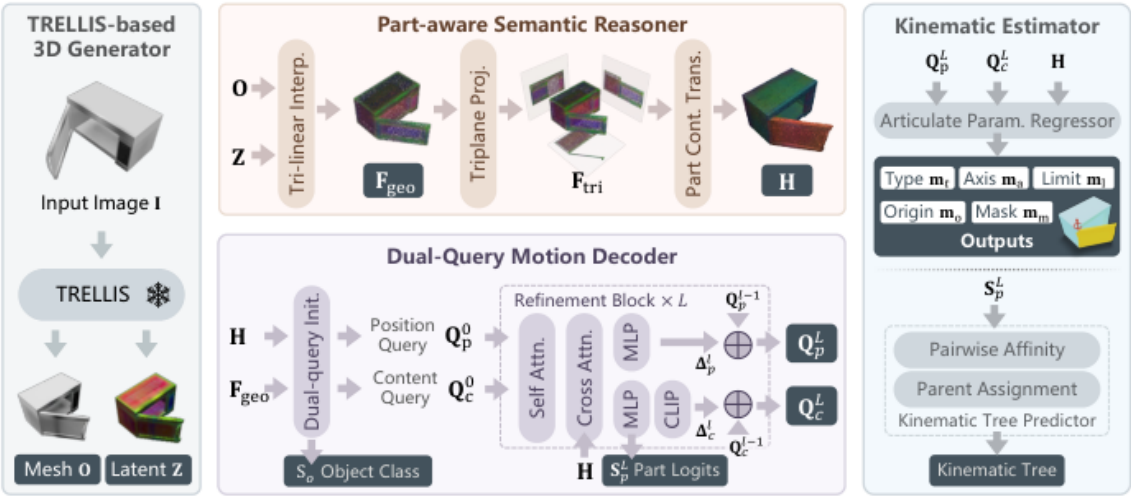

Fig. 2: Overview of MonoArt. TRELLIS-based 3D Generator reconstructs a canonical shape from a single image. Part-Aware Semantic Reasoner derives tri-plane-based part embeddings. Dual-Query Motion Decoder performs iterative motion reasoning, and Kinematic Estimator predicts part-level articulation parameters (motion type, origin, axis, limits) and infers the kinematic tree structure. Note that "Attn.", "Interp.", "Proj.", "Cont.", "Trans.", and "Init." represent "Attention", "Interpolation", "Projection", "Contrast", "Transformer", and "Initialization", respectively. $\oplus$ and $\otimes$ denote element-wise addition and matrix multiplication, respectively.

Tri-linear Interpolation. We sample M point sets $\{\mathbf{p}_m\}_{m=1}^{M}$ on the surface of $\mathbf{O}$, where $\mathbf{p}_m \in \mathbb{R}^3$. Given the sparse voxel feature volume $\mathbf{Z}$, we obtain point-aligned features by tri-linear interpolation over the neighboring voxels:

$$\mathbf{f}_m = \text{TrilinearInterp}(\mathbf{Z}, \mathbf{p}_m), \qquad \mathbf{F}_{\text{geo}} = \{\mathbf{f}_m\}_{m=1}^{M}. \tag{1}$$

Here, each point $\mathbf{p}_m$ is first mapped to its corresponding continuous voxel coordinate in the $N_z \times N_z \times N_z$ grid. The feature $\mathbf{f}_m \in \mathbb{R}^{d_1}$ is then computed as the weighted combination of the eight neighboring voxel features according to standard tri-linear interpolation [56]. This converts the discrete voxel latent $\mathbf{Z}$ into continuous surface-aligned features $\mathbf{F}_{\text{geo}} \in \mathbb{R}^{M \times d_1}$.

Triplane Projection. To incorporate global spatial context while preserving geometric structure, we project the geometry-aligned point features $\mathbf{F}_{\text{geo}}$ onto three orthogonal planes (XY, YZ, ZX) following the triplane formulation [5]. Specifically, each surface point $\mathbf{p}_m$ is orthographically projected onto the three planes according to its 3D coordinates, and its feature $\mathbf{f}_m$ is accumulated to the corresponding 2D grid locations via bilinear interpolation. This yields three feature maps of resolution $N_t \times N_t$, forming $\mathbf{F}_{\text{tri}} \in \mathbb{R}^{3 \times N_t \times N_t \times d_1}$.

Part Contrast Transformer. The triplane features $\mathbf{F}_{\text{tri}}$ are flattened into a token sequence $\mathbf{T} \in \mathbb{R}^{3N_t^2 \times d_1}$ and processed by a multi-layer self-attention Transformer to capture global interactions across planes, producing refined tokens $\mathbf{T}'$. The refined tokens are reshaped back into triplane feature maps $\mathbf{F}'_{\text{tri}} \in \mathbb{R}^{3 \times N_t \times N_t \times d_1}$. For each surface point $\mathbf{p}_m$, we project its 3D coordinate onto the three orthogonal planes and query the refined triplane feature maps via a

Figure 2: Shows the complete MonoArt architecture with all four components and data flow from input image to articulated 3D

reconstruction, illustrating the progressive structural reasoning pipeline.

Limitations

- Struggles with very small parts attached to large objects due to uniform point sampling leading to sparse coverage and less distinctive features
- Relies on learned structural priors over part-whole relationships which may not generalize to novel topologies or uncommon articulation patterns
- Motion parameters can be less accurate for unseen object configurations even when part segmentation remains reasonable
- Limited to objects that can be effectively reconstructed by the frozen TRELLIS backbone